

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully. For each two questions correct, you will receive one course point for a total of two course points.

Question 1: FUNCTIONS

Write a function that performs the calculation: $z = 2x^3 + 3xy + y^2 + 2$. Include an example function call to demonstrate the function in operation.

[illegible]

Question 2: LOOPS

Create a while or repeat loop that runs a time clock and prints each value of the clock in steps of 0.01 until time reaches 10.0.

[illegible]

Question 3: CONDITIONALS

Write a conditional that will sort label a set of 100 random numbers generated by `runif(100)` either “high” if the number is above 0.5 or “low” if it is below.

[illegible]

Question 4: COMBINATIONS

In the `warpbreaks` data set, there are three columns named `breaks` (the number of times the yarn broke), `wool` (the type of wool, either A or B), and `tension` (the tension of the yarn, either L, M, or H). The `wool` and `tension` columns are factored data. Create an empty matrix (one filled with only NA's) that has two rows and three columns, corresponding to the two types of wool and the three possible tensions, respectively. Using two loops, fill in the matrix with the mean number of breaks for each combination of wool and tension options.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]